Capstone Project Part 2

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1-ETL Pipeline in BigQuery

1.1 Extract

The first step in the ETL Pipeline is to extract the data from the source. We received 4 tables in excel format from the team (customers, products, stores, transactions). Since BigQuery dataset section doesn't allow us to upload excel format, we download each table in csv format and then upload it in the dataset section and name it "megamart" The dataset was created under the project ID: 'project-8c0e6044-2686-4c8b-93a'.

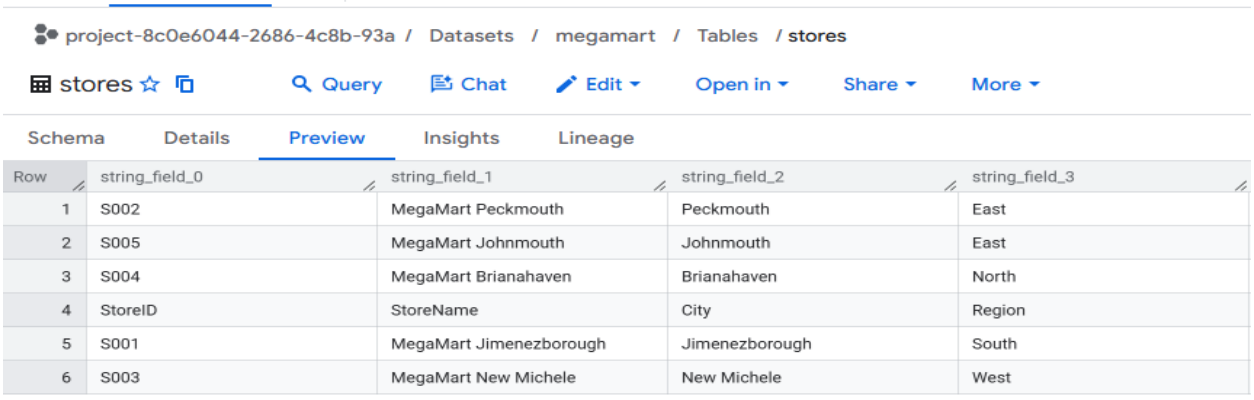
1.2 Transform

To transform the data, we must model it. As asked in the capstone project, we build a star schema in BigQuery using the provided tables.

To create the start schema, we should first create dimension tables then fact table.

- **Create dimension store table**

When we uploaded the store table it displayed like below:



Row	string_field_0	string_field_1	string_field_2	string_field_3
1	S002	MegaMart Peckmouth	Peckmouth	East
2	S005	MegaMart Johnmouth	Johnmouth	East
3	S004	MegaMart Brianahaven	Brianahaven	North
4	StoreID	StoreName	City	Region
5	S001	MegaMart Jimenezborough	Jimenezborough	South
6	S003	MegaMart New Michele	New Michele	West

SQL query to create store dimension table with correct header.

```
CREATE TABLE `project-8c0e6044-2686-4c8b-93a.megamart.dim_stores` AS
SELECT
  string_field_0 AS StoreID,
  string_field_1 AS StoreName,
```

```
  string_field_2 AS City,
  string_field_3 AS Region
FROM `megamart.stores`
WHERE string_field_0 != 'StoreID';
```

➢ Create dimension Customer table

To create dimension customer table, we excluded sensitive information such as name and birth date and we only kept columns required by stakeholders such as Age, Age group and Customer Tenure Months.

- **Customer Tenure** = Current date- Join date
- **Age**=Current Date- Birth date
- **Age group** binning was processed through the following five-step methodology:
  Step 1: Identify Min and Max Age: Profiled the customer dataset to determine the statistical range of customer ages (minimum: 18, maximum: 65+).

```
SELECT
MIN(DATE_DIFF(CURRENT_DATE(), BirthDate, YEAR)) AS min_age,

  MAX(DATE_DIFF(CURRENT_DATE(), BirthDate, YEAR)) AS max_age,

  ROUND(AVG(DATE_DIFF(CURRENT_DATE(), BirthDate, YEAR)), 1) AS avg_age

 FROM `project-8c0e6044-2686-4c8b-93a.megamart.customers`;
```

Job information | Results | Visualization | JSON

Row	min_age	max_age	avg_age
1	19	72	45.7

Step 2: Determine Bin Width and Interval Strategy: Established demographic interval widths (aligning with retail industry standards) to ensure cohorts are intuitive and analytically meaningful for marketing analysis.

```
SELECT
  -- Percentile distribution to see the data spread
  APPROX_QUANTILES(DATE_DIFF(CURRENT_DATE(), BirthDate, YEAR), 4)[OFFSET(1)] AS
p25_age,
  APPROX_QUANTILES(DATE_DIFF(CURRENT_DATE(), BirthDate, YEAR), 4)[OFFSET(2)] AS
median_age,
  APPROX_QUANTILES(DATE_DIFF(CURRENT_DATE(), BirthDate, YEAR), 4)[OFFSET(3)] AS
p75_age
FROM `project-8c0e6044-2686-4c8b-93a.megamart.customers`;
```

Row	p25_age	median_age	p75_age
1	33	46	60

Step 3: Define Boundaries and Validate Distribution: Standard retail lifecycle brackets were selected **(18–25, 26–35, 36–50, 51–65, 65+)** to align with common marketing

consumer stages. A quartile distribution query was run to validate that customers were distributed evenly across these stages rather than heavily clustered in a single group: 25th Percentile: 33 years / Median: 46 years/ 75th Percentile: 60 years.

- **Step 4: Implement SQL Logic:** Executed the conditional transformation using a `CASE WHEN` statement in BigQuery to classify each customer dynamically based on their calculated age.
- **Step 5: Validate and Audit Distribution:** Ran aggregate validation checks (`GROUP BY AgeGroup`) to confirm zero null categories, ensure all customer records were successfully binned, and verify balanced distribution across cohorts.

```
CREATE OR REPLACE TABLE `megamart.dim_customer` AS
SELECT
  CustomerID,
  Gender,
  City,
  JoinDate,

  -- Calculated Fields
  DATE_DIFF(CURRENT_DATE(), BirthDate, YEAR) AS CustomerAge,
  DATE_DIFF(CURRENT_DATE(), JoinDate, MONTH) AS CustomerTenureMonths,

  -- Age Group Binning
 CASE
    WHEN DATE_DIFF(CURRENT_DATE(), BirthDate, YEAR) BETWEEN 18 AND 25 THEN '18-25'
    WHEN DATE_DIFF(CURRENT_DATE(), BirthDate, YEAR) BETWEEN 26 AND 35 THEN '26-35'
    WHEN DATE_DIFF(CURRENT_DATE(), BirthDate, YEAR) BETWEEN 36 AND 50 THEN '36-50'
    WHEN DATE_DIFF(CURRENT_DATE(), BirthDate, YEAR) BETWEEN 51 AND 65 THEN '51-65'
    ELSE '65+'
  END AS AgeGroup

FROM `project-8c0e6044-2686-4c8b-93a.megamart.customers`;
```

➢ Create dimension Products table

```
CREATE OR REPLACE TABLE `project-8c0e6044-2686-4c8b-93a.megamart.dim_products` AS
SELECT
  ProductID,
  ProductName,
  Category,
  SubCategory,
  UnitPrice,
  CostPrice
FROM `project-8c0e6044-2686-4c8b-93a.megamart.products`;
```

➢ Create fact table

```
CREATE TABLE `project-8c0e6044-2686-4c8b-93a.megamart.fact_transactions` AS
```

```sql
SELECT
  -- Primary & Foreign Keys
  t.TransactionID,
  t.CustomerID,
  t.ProductID,
  t.StoreID,

  -- Attributes & Dates
  PARSE_DATE('%Y-%m-%d', CAST(t.Date AS STRING)) AS TransactionDate,
  t.PaymentMethod,
    -- Raw Metrics
  t.Quantity,
  t.Discount,
  p.UnitPrice,
  p.CostPrice,
   -- Required Calculated Fields
  ROUND(t.Quantity * p.UnitPrice * (1 - t.Discount), 2) AS TotalRevenue,
  ROUND(t.Quantity * p.CostPrice, 2) AS TotalCost,
  ROUND((t.Quantity * p.UnitPrice * (1 - t.Discount)) - (t.Quantity * p.CostPrice), 2)
AS TotalProfit,
  ROUND((p.UnitPrice * (1 - t.Discount)) - p.CostPrice, 2) AS ProfitPerUnit

FROM `megamart.transactions` AS t
LEFT JOIN `megamart.dim_products` AS p
  ON t.ProductID = p.ProductID;
```

➢ Data cleaning and validation

- **Duplicate verification:** There is no duplicate found in all the tables

```sql
-- Dimension Stores table
SELECT
StoreID,
COUNT(*)
FROM `project-8c0e6044-2686-4c8b-93a.megamart.dim_stores`
GROUP BY StoreID
HAVING COUNT(*) > 1;
-- Dimension Products table
SELECT
  ProductID,
  COUNT(*)
FROM `project-8c0e6044-2686-4c8b-93a.megamart.dim_products`
GROUP BY ProductID
HAVING COUNT(*) > 1;
-- Dimension Customer table
SELECT
  CustomerID,
  COUNT(*) AS occurrence_count
FROM `project-8c0e6044-2686-4c8b-93a.megamart.dim_customer`
GROUP BY CustomerID
HAVING COUNT(*) > 1;

-- Fact table
```

```sql
SELECT TransactionID,
COUNT(*) AS count
FROM `project-8c0e6044-2686-4c8b-93a.megamart.fact_transactions`
 GROUP BY TransactionID
 HAVING count > 1
```

- **Missing value:** There is no missing value found in all the tables

```sql
-- Fact table
SELECT
 COUNT(*) AS total_rows,
  -- Primary & Foreign Keys
  COUNTIF(TransactionID IS NULL) AS null_transaction_id,
  COUNTIF(CustomerID IS NULL)    AS null_customer_id,
  COUNTIF(ProductID IS NULL)     AS null_product_id,
  COUNTIF(StoreID IS NULL)       AS null_store_id,
    -- Date & Attributes
  COUNTIF(TransactionDate IS NULL) AS null_date,
  COUNTIF(PaymentMethod IS NULL)   AS null_payment_method,
    -- Metrics & Pricing
  COUNTIF(Quantity IS NULL)      AS null_quantity,
  COUNTIF(Discount IS NULL)      AS null_discount,
  COUNTIF(UnitPrice IS NULL)     AS null_unit_price,
  COUNTIF(CostPrice IS NULL)     AS null_cost_price,
    -- Calculated Fields
  COUNTIF(TotalRevenue IS NULL)  AS null_total_revenue,
  COUNTIF(TotalCost IS NULL)     AS null_total_cost,
  COUNTIF(TotalProfit IS NULL)   AS null_total_profit,
  COUNTIF(ProfitPerUnit IS NULL) AS null_profit_per_unit
FROM `project-8c0e6044-2686-4c8b-93a.megamart.fact_transactions`


-- Dimension Customer table
SELECT
COUNT(*) AS total_rows,
COUNTIF(CustomerID IS NULL) AS null_customer_id,
COUNTIF(Gender IS NULL) AS null_gender,
COUNTIF(City IS NULL) AS null_city,
COUNTIF(AgeGroup IS NULL) AS null_age_group,
COUNTIF(CustomerAge IS NULL) AS null_customer_age,
COUNTIF(CustomerTenureMonths IS NULL) AS null_customer_tenure_months,
COUNTIF(JoinDate IS NULL) AS null_JoinDate
 FROM `project-8c0e6044-2686-4c8b-93a.megamart.dim_customer`
-- Dimension Products table
SELECT
 COUNT(*) AS total_rows,
COUNTIF(ProductID IS NULL) AS null_ProductID,
COUNTIF(ProductName IS NULL) AS null_ProductName,
COUNTIF(Category IS NULL) AS null_Category,
COUNTIF(SubCategory IS NULL) AS null_SubCategory,
COUNTIF(UnitPrice IS NULL) AS null_UnitPrice,
COUNTIF(CostPrice IS NULL) AS null_CostPrice,
```

```
 FROM `project-8c0e6044-2686-4c8b-93a.megamart.dim_products`
-- Dimension Stores table
SELECT COUNT(*) as totalrow,
COUNTIF(StoreID IS NULL) AS null_StoreID,
COUNTIF(StoreName IS NULL) AS null_StoreName,
COUNTIF(City IS NULL) AS null_City,
COUNTIF(Region IS NULL) AS null_Region
FROM `project-8c0e6044-2686-4c8b-93a.megamart.dim_stores`
```

- **Fact table integrity**

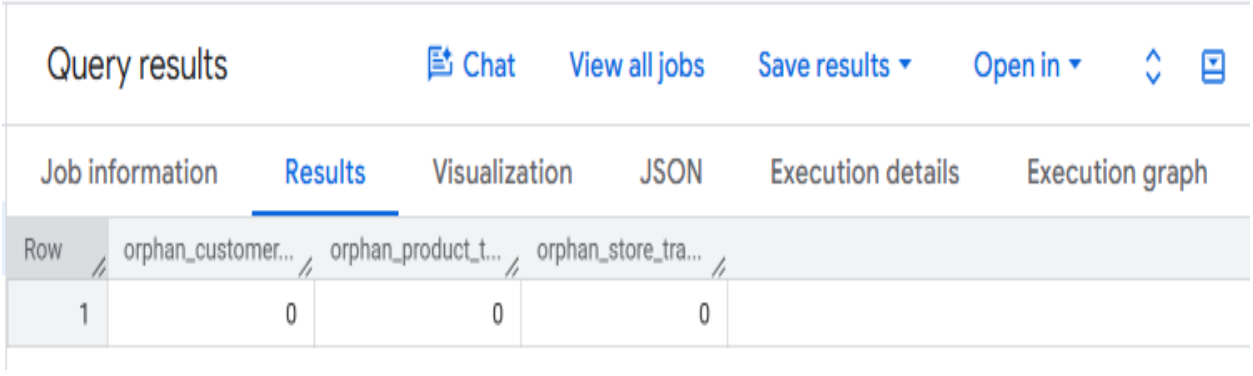
Before joining the tables to create required charts and build the dashboard, verifying that all foreign ID in the fact table exist in dimension table is needed.

```
SELECT
  -- Check for transactions with non-existent customers
  COUNTIF(c.CustomerID IS NULL) AS orphan_customer_transactions,

  -- Check for transactions with non-existent products
  COUNTIF(p.ProductID IS NULL)  AS orphan_product_transactions,

  -- Check for transactions with non-existent stores
  COUNTIF(s.StoreID IS NULL)    AS orphan_store_transactions

FROM `project-8c0e6044-2686-4c8b-93a.megamart.fact_transactions` AS f
LEFT JOIN `project-8c0e6044-2686-4c8b-93a.megamart.dim_customer` AS c ON f.CustomerID
= c.CustomerID
LEFT JOIN `project-8c0e6044-2686-4c8b-93a.megamart.dim_products`  AS p ON f.ProductID
= p.ProductID
LEFT JOIN `project-8c0e6044-2686-4c8b-93a.megamart.dim_stores`    AS s ON f.StoreID
= s.StoreID;
```



1.3 Load

- **Analytical Reporting Layer:** joining fact and dimension tables into a single source of truth for BI.

```
CREATE OR REPLACE VIEW `megamart.vw_sales_reporting` AS
SELECT
  -- Fact Keys & Date
  f.TransactionID,
  f.TransactionDate,
  EXTRACT(YEAR FROM f.TransactionDate) AS TransactionYear,
  EXTRACT(MONTH FROM f.TransactionDate) AS TransactionMonth,
  f.PaymentMethod,

  -- Customer Demographics (PII-Minimised: No names or exact birthdates)
  c.CustomerID,
  c.Gender,
  c.City AS CustomerCity,
  c.CustomerAge,
  c.AgeGroup,
  c.CustomerTenureMonths,

  -- Product Attributes
  p.ProductID,
  p.ProductName,
  p.Category,
  p.SubCategory,

  -- Store / Location
  s.StoreID,
  s.StoreName,
  s.City AS StoreCity,
  s.Region AS StoreRegion,

  -- Core Financial Metrics
  f.Quantity,
  f.UnitPrice,
  f.CostPrice,
  f.Discount,
  f.TotalRevenue,
  f.TotalCost,
  f.TotalProfit,
  f.ProfitPerUnit

FROM `project-8c0e6044-2686-4c8b-93a.megamart.fact_transactions` AS f
INNER JOIN `project-8c0e6044-2686-4c8b-93a.megamart.dim_customer` AS c ON f.CustomerID
= c.CustomerID
INNER JOIN `project-8c0e6044-2686-4c8b-93a.megamart.dim_products`  AS p ON f.ProductID
= p.ProductID
```

```
INNER JOIN `project-8c0e6044-2686-4c8b-93a.megamart.dim_stores`    AS s ON f.StoreID
= s.StoreID;
```

- **Final Verification & KPIs Calculation:** A summary reconciliation query on `vw_sales_reporting` confirming total transactions, total revenue, and overall margins
- **KPI calculation**

```
SELECT
  -- Transaction & Customer Volume
  COUNT(*) AS total_transactions,
  COUNT(DISTINCT CustomerID) AS active_customers,
  SUM(Quantity) AS total_units_sold,

  -- Financial Metrics
  ROUND(SUM(TotalRevenue), 2) AS total_revenue,
  ROUND(SUM(TotalCost), 2) AS total_cost,
  ROUND(SUM(TotalProfit), 2) AS total_profit,

  -- Margin & Order Performance
  ROUND(SUM(TotalProfit) / SUM(TotalRevenue) * 100, 2) AS overall_profit_margin_pct,
ROUND(SUM(TotalRevenue) / COUNT(*), 2) AS average_order_value

FROM `project-8c0e6044-2686-4c8b-93a.megamart.vw_sales_reporting`;
```

Metric	Baseline Value	Description
Total Transactions	**5,000**	Total orders processed
Active Customers	**200**	Total unique purchasing customers across all transactions
Total Units Sold	**14,949**	Total volume of physical items sold
Total Revenue	**$14,301,903.79**	Gross sales revenue after customer discounts
Total Cost	**$10,475,588.48**	Direct cost of goods sold (COGS)
Total Profit	**$3,826,314.91**	Net gross operating profit generated
Overall Profit Margin	**26.75%**	Portfolio-wide gross margin percentage
Average Order Value (AOV)	**$2,860.38**	Mean transaction revenue per checkout

- **Customer age and demographic distribution insight**

```sql
SELECT
  AgeGroup,
  COUNT(*) AS customer_count,
  ROUND(COUNT(*) * 100.0 / SUM(COUNT(*)) OVER(), 1) AS percentage_share
 FROM `project-8c0e6044-2686-4c8b-93a.megamart.vw_sales_reporting`
GROUP BY AgeGroup
ORDER BY AgeGroup;
```

AgeGroup	Customer_count	Percentage_share (%)
18–25	676	13.5%
26–35	837	16.7%
36–50	1,356	27.1%
51–65	1,534	30.7%
65+	597	11.9%